

14.3 Partial Derivatives

1 Warm Up

Question: What does the derivative of a function of one variable represent?

2 Motivation

For a function of one variable (two dimensions on the plane $f(x)$) we have an unambiguous definition of a derivative. In 2D the derivative at a point x is the slope of the tangent line to the curve at that point. The issue when we talk about surfaces in 3D is that there are infinite tangent lines to a surface at a point (x, y) what do we do here? In this chapter we begin to answer this question.

3 Definitions & Theorems

1. Let $f(x, y)$ be a function of two variables aka a surface. Since there are infinite tangent lines to the surface at (x, y) , we specifically set our direction to get one and only one line. Two directions which make sense to select is the x direction and the y direction.
2. Since we want our tangent line to be in the direction of the x -axis, it must lie in a plane parallel to the xz plane and cuts through our surface creating a curve. Any plane parallel to the xz plane has y constant. Thus, we call the derivative in the x direction the partial derivative with respect to x and we denote this f_x or $\partial f / \partial x$. We think about y in this case as a constant.
3. Similarly the partial derivative with respect to y denoted f_y or $\partial f / \partial y$ is the tangent line to our surface in the direction of the y -axis. Here our line lies in a plane parallel to the yz plane and we consider x as a constant.
4. We can also write these derivatives in terms of their limit definitions as follows.

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}.$$

5. This definition can easily be extended beyond functions of two variable for instance if $w = f(x, y, z)$ is a function in $\mathbb{R}^4$ we have,

$$\partial w / \partial x = f_x(x, y, z) = \lim_{h \rightarrow 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}$$

6. We of course can take higher derivatives than just the first partial derivatives. The following figure

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

7.

Theorem 1. (*Clairaut's Theorem*) If f is defined on a disk D which contains (a, b) and f_x and f_y are both continuous on D then $f_{xy}(a, b) = f_{yx}(a, b)$.

4 Examples

See partial-derivative worksheet

5 Scratch Work